Show that the argument form with premises $p_1, p_2, \ldots, p_n$ and conclusion $q \rightarrow r$ is valid if the argument form with premises $p_1, p_2, \ldots, p_n, q$, and conclusion r is valid.

$\equiv (P_1 \wedge P_2 \wedge \ldots \wedge P_n) \longrightarrow (q \rightarrow r)$

$\equiv \neg(P_1 \wedge P_2 \wedge \ldots \wedge P_n) \vee (\neg q \vee r)$

$\equiv \neg(P_1 \wedge P_2 \wedge \ldots \wedge P_n \wedge q) \vee r$

$\equiv P_1 \wedge P_2 \wedge \ldots \wedge P_n \wedge q \longrightarrow r$

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For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises. 1/10/2023 14:59

a) "If I play hockey, then I am sore the next day." "I use the whirlpool if I am sore." "I did not use the whirlpool."

b) "If I work, it is either sunny or partly sunny." "I worked last Monday or I worked last Friday." "It was not sunny on Tuesday." "It was not partly sunny on Friday."

c) "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragonflies."

d) "Every student has an Internet account." "Homer does not have an Internet account." "Maggie has an Internet account."

e) "All foods that are healthy to eat do not taste good." "Tofu is healthy to eat." "You only eat what tastes good." "You do not eat tofu." "Cheeseburgers are not healthy to eat."

f) "I am either dreaming or hallucinating." "I am not dreaming." "If I am hallucinating, I see elephants running down the road."

a) P I play hockey | $P \rightarrow q$, $q \rightarrow s$ } $P \rightarrow s$ $\equiv \neg s \rightarrow \neg p$

a) p I play hockey
q I am sore
s I use whirlpool

$p \to q$, $q \to s$, $\neg s$ $\quad \equiv \quad p \to s \equiv \neg s \to \neg p$

$\neg p$

---

b) P(x) I work on x
q it's sunny
s it's partly sunny

$\forall x \; P(x) \to [q(x) \vee s(x)]$
$P(Mon) \vee P(Fri)$
$\neg q(Tue)$
$\neg s(Fri)$

$P(Mon) \to [q(Mon) \vee s(Mon)]$
$P(Mon) \vee P(Fri)$

$\neg q(Mon) \wedge \neg s(Mon) \to \neg p(Mon)$
$\neg p(Mon) \to P(Fri)$

$\neg q(Mon) \wedge \neg s(Mon) \to P(Fri)$

---

c) P(x): x is an insect
q: x has six legs
s(x,y): x eats y

$\forall x [P(x) \to q(x)]$
$P(D)$
$\neg q(SP)$
$S(SP, D)$

$q(D) \wedge \neg P(SP)$

$\exists x, y, (\neg P(x) \wedge P(y)) \to S(x,y)$

Determine whether these are valid arguments

Determine whether these are valid arguments.

a) If x is a positive real number, then x^2 is a positive real number. Therefore, if a^2 is positive, where a is a real number, then a is a positive real number.

b) If $x^2 \neq 0$, where x is a real number, then $x \neq 0$. Let a be a real number with $a^2 \neq 0$; then $a \neq 0$.

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a) $P(x)$: x is positive real

$q(x)$: x^2 is positive real

$P(x) \longrightarrow q(x)$

$q(x) \longrightarrow P(x)$ No?!

$x = -2$

b) $P(x)$: $x^2 \neq 0$ $\quad P(x) \longrightarrow q(x)$

$q(x)$: $x \neq 0$

$P(a) \longrightarrow q(a)$ ✓

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Use rules of inference to show that if $\forall x(P(x) \rightarrow (Q(x) \wedge S(x)))$ and $\forall x(P(x) \wedge R(x))$ are true, then $\forall x(R(x) \wedge S(x))$ is true.

$\forall x \left[P(x) \longrightarrow (Q(x) \wedge S(x))\right]$

$\forall x \left[P(x) \wedge R(x)\right]$

$\forall x \left[R(x) \wedge S(x)\right]$

$P(a) \longrightarrow Q(a) \wedge S(a)$

$P(a) \wedge R(a) \longrightarrow R(a)$

$P(a) \wedge R(a)$

$\downarrow$

$P(a)$

$\downarrow$

$Q(a) \wedge S(a)$

$\downarrow$

$S(a)$

$R(a) \wedge S(a)$

Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$ and $\forall x((\neg P(x) \wedge Q(x)) \rightarrow R(x))$ are true, then $\forall x(\neg R(x) \rightarrow P(x))$ is also true, where the domains of all quantifiers are the same.

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$\forall x (P(x) \vee Q(x))$

$\forall x ([\neg P(x) \wedge Q(x)] \longrightarrow R(x))$

$\forall x [\neg R(x) \longrightarrow P(x)]$

$P(a) \vee Q(a)$

$(\neg P(a) \wedge Q(a)) \longrightarrow R(a) \equiv \neg(\neg P(a) \wedge Q(a)) \vee R(a)$

$\equiv P(a) \vee Q(a) \vee R(a)$

Or Consider $\neg R(a) \equiv T$

$(\neg P(a) \wedge Q(a)) \longrightarrow R(a)$

$\neg R(a)$

$\neg(\neg P(a) \wedge Q(a)) \equiv P(a) \vee \neg Q(a)$

$P(a) \vee \neg Q(a)$ True $\quad P \vee (T, F)$

$P(a) \vee Q(a)$ True

$P(a)$ why ?!

$Q(a) \longrightarrow P(a)$

$\neg Q(a) \longrightarrow P(a)$

$\neg P(a) \longrightarrow Q(a)$

$\neg P(a) \longrightarrow P(a)$

$P(a) \wedge P(a)$

$P(a)$

Use rules of inference to show that if $\forall x(P(x) \vee Q(x))$, $\forall x(\neg Q(x) \vee S(x))$, $\forall x(R(x) \rightarrow \neg S(x))$, and $\exists x \neg P(x)$ are true, then $\exists x \neg R(x)$ is true.

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$\forall x \, (P(x) \vee Q(x))$

$\forall x \, (\neg Q(x) \vee S(x))$

$\forall x \, (R(x) \longrightarrow \neg S(x))$

$\exists x \, \neg P(x)$

$\neg P(c)$

$R(c) \longrightarrow \neg S(c) \equiv S(c) \longrightarrow \neg R(c)$

$R(c) \rightarrow \neg S(c) \equiv S(c) \rightarrow \neg R(c)$

$\neg Q(c) \vee S(c) \equiv Q(c) \rightarrow S(c)$

$P(c) \vee Q(c) \equiv \neg P(c) \rightarrow Q(c)$

$\boxed{\neg P(c) \rightarrow \neg R(c)}$

$\neg R(c) \Rightarrow \exists x \neg R(x)$

Use a direct proof to show that every odd integer is the difference of two squares. [*Hint:* Find the difference of the squares of $k + 1$ and k where k is a positive integer.]

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$x = p^2 - L^2$?!

P, L are integer.

if x is odd. $\Rightarrow x = 2n+1$

---

if P, L are even

$P = 2m$

$L = 2k$

$p^2 - L^2 = 2[2m^2 - 2k^2]$

---

if P, L are odd.

$P = 2m+1$

$L = 2k+1$

$p^2 - L^2 = 4m^2 + 2m + 1 - 4k^2 - 2k - 1$

$= 2[\ -\ ]$ even.

---

without loss of generality

$P = 2m+1$

$L = 2k$

$P = 2m+1$

$L = 2k$

$P^2 - L^2 = 4m^2 + 2m + 1 - 4k^2$

$= 2\underbrace{[2m^2 + m - 2k^2]}_{n} + 1$

$2m^2 + m - 2k^2 = n$

$m = k$

$m = n$

$P = 2m+1$

$L = 2m$

Prove !!

$x = 2n+1 = n^2 - n^2 + 2n + 1$

$= \underbrace{(n+1)^2}_{} - \underbrace{n^2}_{}$

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Prove that if n is a perfect square, then $n + 2$ is not a perfect square.

n is perfect

$n = k^2$ $\quad k = 0, 1, 2, \cdots$

$(k+1)^2$

$k^2 + 2k + 1 = n + 2k + 1$

$k > 0$

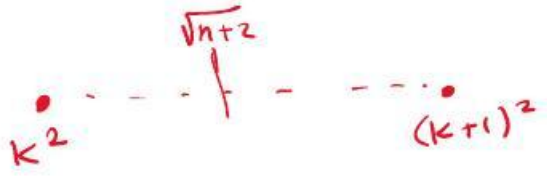
$k=0$

$n=0$

$n+2$ is perfect

$k>0$

$n+2k+1 > n+2$



Prove or disprove that the product of two irrational numbers is irrational.

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wrong !! $\sqrt{3} \times \sqrt{3} = \boxed{3}$ rational

Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

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True $x \in Q'$ $\frac{a}{b} \in Q$

$x\,a/b \in Q'$

proof by contradiction

suppose $x\,a/b \in Q$

$\Rightarrow \left(\frac{b}{a}\right)\left(\frac{x\,a}{b}\right) \in Q$

$\Rightarrow x \in Q$ contradict

Prove that if x is irrational, then $1/x$ is irrational.

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by contrapositive $\frac{1}{x} \in \mathbb{Q} \rightarrow x \in \mathbb{Q}$

$\frac{1}{x} \in \mathbb{Q} \Rightarrow \frac{1}{x} = \frac{a}{b} \quad a, b \in \mathbb{Z}$

$\Rightarrow x = \frac{b}{a} \Rightarrow x \in \mathbb{Q}$

Prove that if m and n are integers and mn is even, then m is even or n is even.

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by contrapositive

n, m are odd $\rightarrow$ mn is odd

$n = 2k+1$
$m = 2l+1$
$\Big\} (2k+1)(2l+1) \rightarrow$ odd

Show that if n is an integer and $n^3 + 5$ is odd, then n is even using

a) a proof by contraposition.
b) a proof by contradiction.

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a) n is odd $\rightarrow$ n^3+5 is even

$n = 2k+1$

$n^3+5 = (2k+1)^3 + 1 = 2^3k^3 + 3 \times 2^2k^2 + 3 \times 2 \times k + \underbrace{1+5}_{6}$

$= 2\left[2^2k^3 + 2 \times 3 \times k^2 + 3 \times k + 3\right]$ ✓

$= 2\left[2^2 k^3 + 2 \times 3 \times k^2 + 3 \times k + 3\right]$ ✓

b) $n^3 + 5$ is odd

n is odd

Show that if you pick three socks from a drawer containing just blue socks and black socks, you must get either a pair of blue socks or a pair of black socks.

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proof by contradict
- you pick 3 socks
- but you did not get any similar pair

⇒ You get one black and one blue and one must be neither a contradiction.

Show that at least ten of any 64 days chosen must fall on the same day of the week.

proof by contradiction.
Suppose we have 9 days fallen in the same week.
⇒ 2 days are repeated.

64 $7 \times 9 = 63$

remainder 1

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. Use a proof by contradiction to show that there is no rational number r for which $r^3 + r + 1 = 0$. [*Hint:* Assume

let $r = a/b$ and $r^3 + r + 1 = 0$

$\left(\frac{a}{b}\right)^3 + \left(\frac{a}{b}\right) + 1 = 0$ $\quad a, b \neq 0$

$a^3 + b^3 + ab^2 = 0$
$= 2(0)$

a/b there is no common divisors

divide by 2

a, b cannot both be even,

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a, b are odd

a^3 is odd

b^2 is odd

sum of these odd will is odd — cannot be even

---

a is odd $\quad b$ is even

a^3 is odd
b^3 is even
ab^2 is even
} odd ✓

a is even $\quad b$ is odd

a^3 is even
b^3 is odd
ab^2 is even
} odd ✓

Prove that if n is a positive integer, then n is odd if and only if $5n + 6$ is odd.

n is odd $\Longleftrightarrow$ $5n+6$ is odd

$\Rightarrow$

$n = 2m+1$

$5n+6 = 5(2m+1)+6 = 10m+5+6 = 10m+11$

$= 2(5m+5)+1$

$\Leftarrow$

$5n+6$ is odd

proof by contrapositive

n is even $\Rightarrow$ $5n+6$ is even

$5(2m)+6 = 2(5m+3)$ #
